

Design a Hardware Applying Fog Removal Algorithm using Median Dark Channel Prior for Autonomous Driving Car

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Abstract— Autonomous cars require camera images to obtain accurate information about the driving environment. However, there are times when clear images cannot be obtained due to fog, rain, or nighttime, which may adversely affect the quality of autonomous driving. In this paper, we propose an algorithm and hardware architecture to remove fog or rain from images in real time. A commonly used method to remove fog is the Dark Channel Prior method, which has the disadvantage of causing blocking phenomenon. Taking this into consideration, we modified it to the Median Dark Channel method. To design this algorithm in Verilog HDL, we could reduce hardware resource by sharing comparators and achieve real-time processing by pipelining and line buffer to increase memory usability.

Keywords— fog removal algorithm, Verilog HDL, Image processing, autonomous driving

I. INTRODUCTION

As industrial technology develops, image processing is becoming indispensable in the era of digital transformation. Representative examples include autonomous cars, CCTV, and smartphones. Autonomous driving functions that improve driver's convenience and safety receive attention and continue to be developed[1]. Among the autonomous driving functions, cruise control uses sensors and cameras to prevent vehicles from leaving their lanes and maintain certain distance from the vehicle in front. And multiple image sensors are also used in the around-view function, which helps the driver's convenience especially when parking a car. The clearer the image, the higher the reliability and the higher the accuracy of image recognition, so research in the field of image quality improvement is being actively conducted. Among many image processing algorithms, the fog removal algorithm[2] is mainly used in CCTV, making it easy to observe and audit road conditions or situations in specific areas. In this paper, the fog removal algorithm is applied to moving cars rather than fixed CCTV. The fog generated in images reduces the contrast and clarity of images, becoming a fatal problem in image processing systems that detect objects or

perform filtering. Therefore, in autonomous driving, it is essential to remove fog that occurs in images. For drivers, it is not easy to recognize lanes on foggy days, so incorporating a fog removal algorithm into the camera above would help with safer driving. The fog removal algorithms include methods for removing fog using multiple images or additional information and methods for removing fog using a single image[3]. Since fog removal using multiple images or additional information requires relatively greater hardware and software costs than the fog removal method using a single image, the fog removal method using a single image is used in this paper. In this paper, the fog removal algorithm is implemented in FPGA and designed in Verilog HDL.

II. REALTED WORK

Scattering of light occurs due to fine particles in the atmosphere such as fog, smoke, or yellow dust. Scattering of light means that light collides with particles in the air and changes its direction. The new light source that occurs at this time is called airlight. The airlight is modeled as , where $A(1 - t(x))$ is the airlight value in the image, and $t(x)$ represents the amount of light that is not scattered through transmission reaching the field of view. Transmission is as in equation (1).

$$t(x) = e^{-\beta d(x)} \quad (1)$$

In equation (1), β is the scattering coefficient of the atmosphere, d represents the distance between the subject and the field of view, and indicates that the light of the subject is attenuated exponentially according to the value of distance d .

Using the airlight modeling, the foggy image is modeled as shown in equation (2).

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (2)$$

In equation (2), $I(x)$ represents the input image, and $J(x)$ represents the image with the fog removed. $J(x)t(x)$ in the first part is the product of the transmission amount $t(x)$ and the fog-

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free image $J(x)$, which represents the image due to the direct attenuation effect without airlight.

III. THE PROPOSED METHOD

A dark channel refers to the set of the smallest RGB values of one pixel within a certain patch size in an image[4][5]. Generally, in images without fog, you can see that one of the RGB values of each pixel is close to 0. If you check the gray level of the dark channel of images without fog, you can obtain images with low brightness, but you can see that images with fog have high overall brightness. Low brightness in the dark channel is mainly caused by shadows, solid-colored objects or surfaces, or dark objects or surfaces[6]. Outdoor images mainly contain a lot of shadows or consist of monochromatic objects, so the brightness of the dark channel is low.

The flow chart of the fog removal algorithm is shown in Fig. 1. The dark channel is obtained from the input foggy image, airlight is found using the dark channel, and the transmission amount is obtained and confirmed. Use this to restore the image.

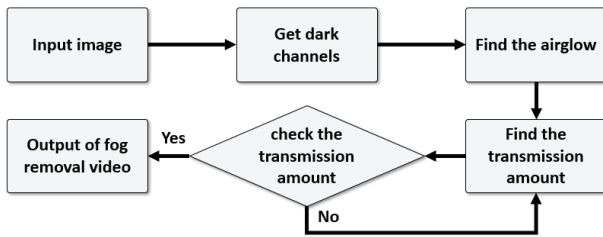


Fig. 1. Flow chart of fog removal algorithm

There is a problem that blocking occurs when restoring an image using a dark channel. The blocking phenomenon must be resolved using a highly complex bilateral Filter or matting Filter. In this paper, we solve this problem by using the characteristics of the median filter to reduce complexity and solve the halo effect when designing with an FPGA. This is called median dark channel prior. Fig. 2 shows the median dark channel period method[7].

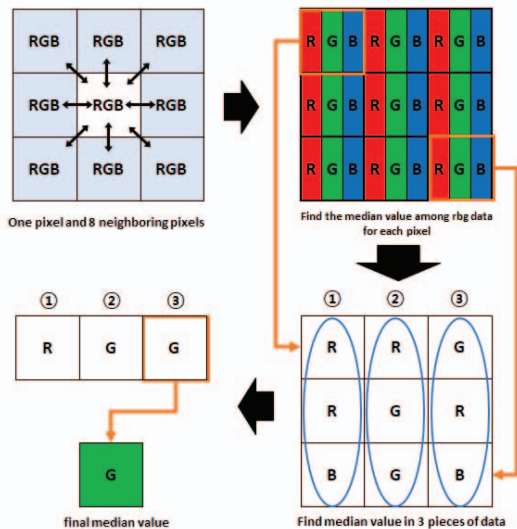


Fig. 2. Finding median value in dark channel

A total of 9 pixel data is stored in the buffer, with 8 neighboring pixels based on one pixel. One pixel's RGB data is 24 bits and has 3 pieces of data of 8 bits each, so 27 pieces of data are stored in the buffer. At each pixel, the median value is selected from the RGB values and stored. Only a total of 9 median data remain, and the median filter is applied to the vertical axis in units of 3 pixels, as shown in Fig. 2. Three median values are obtained, and the final median value from these data is obtained through the median filter once more. In order to proceed with the above process for each of the 3 lines, it has a structure that uses 3 line buffers.

IV. CONCLUSION

In this paper, we proposed a median dark channel before-based fog correction adjustment to suit the hardware design and proposed a method of using a line buffer to enable real-time processing. The characteristics of the median filter were used to solve the block effect and halo effect problems that occurred in the existing dark channel prior. Due to the nature of the sorting algorithm, it consumes a lot of resources, so to solve this problem, we proposed a method to reduce the number of comparators by comparing only three pieces of data. If this method is applied to the fog removal algorithm, it appears that clearer images can be obtained for lane recognition during autonomous driving in cars.

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